Information Fusion for Classification of Hyperspectral and LiDAR Data Using IP-CNN

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***Abstract*— Joint use of multisensor information has attracted considerable attention in the remote sensing community. While applications in land-cover observation benefit from information diversity, multisensor integration technique is confronted with many challenges, including inconsistent size of data, different data structures, uncorrelated physical properties, and scarcity of training data. In this article, an information fusion network, named interleaving perception convolutional neural network (IP-CNN), is proposed for integrating heterogeneous information and improving joint classification performance of hyperspectral image (HSI) and light detection and ranging (LiDAR) data. Specifically, a bidirectional autoencoder is designed to reconstruct hyperspectral and LiDAR data together, and the reconstruction process is trained with no dependence upon annotated information. Both HSI-perception constraint and LiDAR-perception constraint are imposed on multisource structural information integration. Accordingly, fused data are fed into a two-branch CNN for final classification. To validate the effectiveness of the model, the experiments were conducted using three datasets (i.e., Muufl Gulfport data, Trento data, and Houston data). The final results demonstrate that the proposed framework can significantly outperform state-of-the-art methods even with small-size training samples.**

***Index Terms*— Convolutional neural network (CNN), deep learning, hyperspectral image (HSI), joint classification, light detection and ranging (LiDAR) data, pattern recognition, remote sensing.**

I. Introduction

REMOTE sensing classification plays an important role in land-cover observation, environmental science, forest management, and so on [1]–[3]. Meanwhile, recent advances in sensor technology have made remarkable progress, thereby providing more data resources for remote sensing classification applications [4], [5]. Among different data sources, hyperspectral image (HSI) contains hundreds of spectral bands, which provides detailed spectral information of land covers [6], whereas light detection and ranging (LiDAR) data can capture elevation distribution about the surveyed area, which is related to height, size, and shape of specific objects even under cloud-cover conditions [7], [8]. The diagnostic spectral information of HSI enables fine-grained classification of observed objects, whereas the vertical structural information of LiDAR can be applied for obscuration (cloud/shadow) area exploration [7], [8]. Thus, joint use of HSI and LiDAR data has aroused considerable consideration and has been widely used in remote sensing tasks to provide comprehensive interpretation for the study area [9], [10].

As HSI is the data source with abundant spectral information, accurate analysis of HSI can help improve the joint classification performance for HSI and LiDAR data. Great efforts have been made to developing effective HSI classification technique in remote sensing community [11]–[18]. For example, support vector machine (SVM) [19] and kernel extreme learning machine (ELM) [20] have been applied for HSI classification. However, simple classification strategy ignores the use of spatial correlation among pixels, limiting the classification performance. Wang *et al.* [13] further presented locality constraint and structure-preserving strategy for HSI classification. In addition to the conventional methods, deep learning-based works arouse wide attention for their capabilities of automatically extracting robust and high-level features. HSI classification algorithms based on convolutional neural network (CNN) have been extensively studied, e.g., CNN-based pixel-pair framework (CNN-PPF) [14] and convolutional recurrent neural network (CRNN) [16]. The utilization of spatial information is also important for deep networks. In context CNN [15], spatial and spectral features are extracted in parallel, and the final classification result can be obtained by using vectorized features. In [17] and [18], sequential frameworks were adopted to automatically learn spatial and spectral features. The fast dense spectral–spatial convolution network (FDSSCN) [17] was also demonstrated to be effective in single-HSI classification. However, these methods are not customized for multisource classification methods. Hence, although these methods show excellent performance in HSI classification, it still may fail in multisource data joint classification.

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On the other hand, some feature-fusion- and decision-fusion-based classification methods were developed for HSI and LiDAR data joint classification. For example, in [21], morphological operators were exploited to extract both hyperspectral and LiDAR features. However, the features learned in morphological algorithms are source-related, and simple concatenation of features may contain redundant information. Also, multisource information cannot be equally represented when HSI and LiDAR features were used in a concatenation manner [22], limiting overall classification performance. Thus, in [23] and [24], graph-based feature-fusion techniques were proposed for joint classification of HSI and LiDAR data. Xu *et al.* [25] then proposed a stacking-and-mapping way to handle the multisource features representation problem, and the two-branch architecture was demonstrated to be efficient. Furthermore, architectures of multibranch network with postprocessing and semisupervised settings were then developed [26], [27], and these methods have been validated on the data provided by the Image Analysis and Data Fusion Technical Committee in the 2018 IEEE GRSS Data Fusion Contest [28]. In particular, to solve the problem caused by unbalanced features, decision-fusion methods for HSI and LiDAR data classification have also been presented [29], [30] to skip the step involved with feature fusion. Feature-fusion strategy and decision-fusion strategy were also simultaneously used [31], [32] to improve the classification performance recently. Although these feature- and decision-fusion methods have been validated to be effective in joint classification, their fusion and classification strategies are semantic-related and task-specific, where both heterogeneous properties and complementary information between HSI and LiDAR data cannot be properly parsed only with the guidance of annotation samples.

Due to heterogeneous properties of different data sources, the representations of objects show great differences between HSI and LiDAR data, including data size and data structure. Therefore, for multiple sources, it is difficult for a direct-classification model to yield satisfactory joint analysis performance based on limited training data. To tackle this issue, unsupervised CNN models were proposed [33], [34] based on the encoder–decoder architecture, and the data reconstruction strategy is demonstrated to be capable of better activating the neurons across modalities [35]. Although the use of unlabeled samples is a feasible solution to improve the joint classification performance, heterogeneous and complementary information is still not efficiently explored in these unidirectional mapping designs [33]. How to effectively establish the complementary connections between multisources is still an open question. In remote sensing data, the utilization of structure information is of great importance [36], and the key to construct complementary connections of different sources is to build structure relationships between sources. In terms of structure relationship, Guo *et al.* [37] used the second-order interaction of Gram matrix to control the structure difference between the clean and degraded images. In particular, Gram matrices matching of neural activations is shown to be an effective way for heterogeneous distribution alignment between the images [38]. Inspired from these works, the idea of Gram matrix matching of neural activations is modified in this article to maintain the complementary structure within aligned distributions of HSI and LiDAR data.

In this article, we propose a structural information fusion framework for HSI and LiDAR data joint classification. The information fusion network, named interleaving perception CNN (IP-CNN), is designed for integrating complementary information and alleviating the complexity of heterogeneous information regulation for the following classification. Specifically, the IP-CNN is composed of two intersecting branches (HSI and LiDAR data reconstruction branches), and a complementary-structure control (CSC) module is placed at the intersection of the two reconstruction branches. After data-driven fusion preprocessing, the last intersection layer of IP-CNN is connected to the two-branch CNN to obtain the final classification result. To sum up, the main contributions of this article are as follows.

1) An IP-CNN is proposed for complementary information integration and heterogeneous information control. In particular, for the first time, we design two types of multisource structure constraints for HSI and LiDAR characteristic integration, and Gram matrices are utilized in multisource complementary information preservation, to promote structural information extraction.
2) In the hybrid joint classification framework, data-driven analysis and task-specific adaption are embedded, enabling parsing multiple information in an orderly fashion. Such design facilitates the full use of multisource properties and semantic information, achieving the improved classification performance.
3) The neural weights learning process of the designed IP-CNN is completely unsupervised, which is independent of labeled samples. In doing so, the task-specific classification procedure can be well generated to a supervised manner. Compared with direct-classification models, the proposed framework achieves substantially improved classification performance under different training data sizes.

The remainder of this article is organized as follows. The proposed framework is introduced in Section II. The experimental results and analysis are presented in Section III. Finally, Section IV summarizes with concluding remarks.

II. Proposed Classification Framework

An overview of the proposed framework is shown in Fig. 1, which includes a data-driven complementary information fusion network and a semantic-oriented transferring network.

A. Complementary Information Fusion Based on IP-CNN

The interleaving perception network consists of two reconstruction branches and a CSC module, as shown in Fig. 1. More details of the IP-CNN are elaborated in Fig. 2(a).

The front part of IP-CNN serves as the aggregation module, which begins by using 3×3 convolution operation to extract features from multiple inputs separately. The number of feature-map channels is D in each layer of the aggregation procedure, and the structural operations beyond two branches

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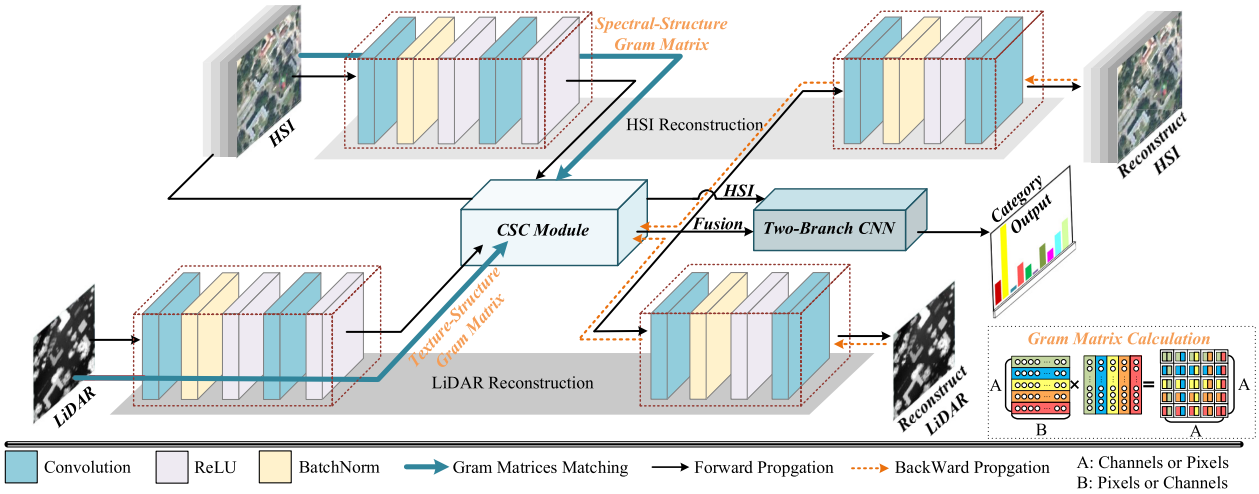


Fig. 1. Illustration of the proposed joint classification architecture.

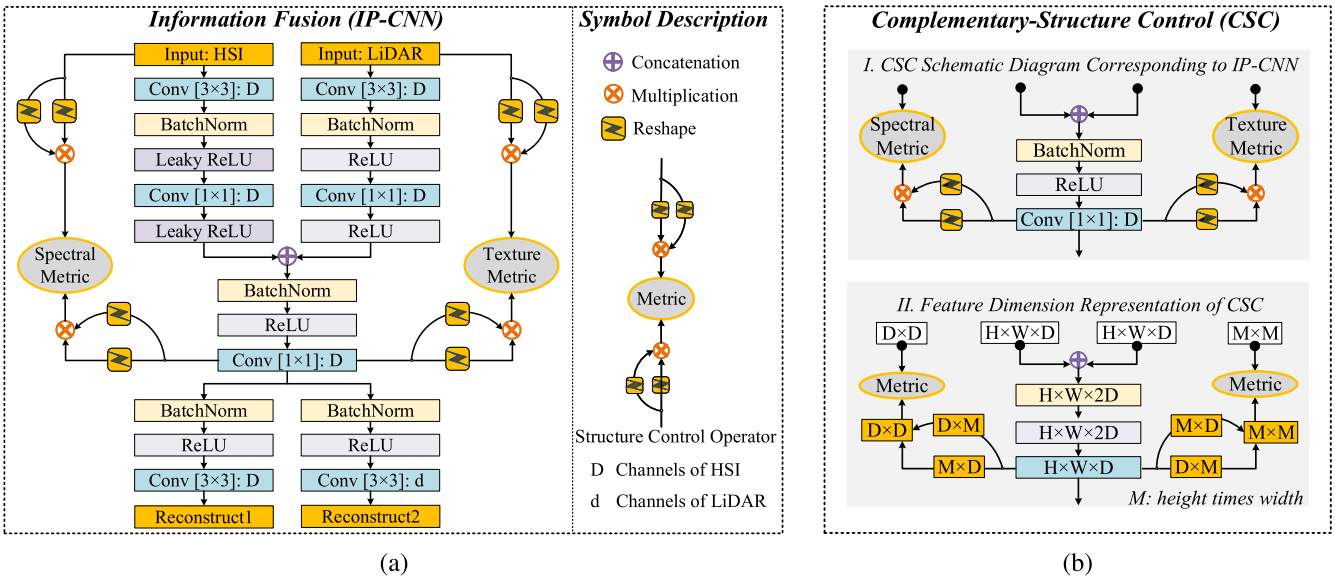


Fig. 2. Architecture of the IP-CNN: (a) parameter-setting for IP-CNN and (b) detailed setting for CSC module in IP-CNN.

are consistent with each other, while in the intersection part, hidden features derived from two branches are combined to form a fusion set, and the fusion set is then fed into the following refinement process to generate the complementary fusion block based on the CSC module. An illustration of the CSC module is shown in Fig. 2(b). Here, the idea of second-order interaction alignment of Gram matrix is modified to maintain the complementary structure of the fusion block [37], [39], [40]: the Gram matrix from LiDAR source can be considered as the texture-structure reference for fused features, and the Gram matrix from HSI source can be considered as the spectral-structure reference for fused features. In this way, joint control of the Gram matrices between input sources and fused features can promote the preservation of complementary structure. The reconstruction setting can help to maintain the integrity of the fused information.

Let $H = [H_1, \ldots, H_c] \in R^{M \times D}$ denote an HSI dataset with M samples over C classes, with D the spectral dimensions per sample. Let $L = [L_1, \ldots, L_c] \in R^{M \times d}$ denote LiDAR data with feature dimension of d. Besides, $\phi(\cdot)$ is defined as a composite function of consecutive operations in one hidden layer. Accordingly, $f^{(l)}(H, L) = \phi^{(l)}(\cdots(\phi^{(1)}(H, L)))$ denotes the hidden features obtained in the lth layer and $\hat{d}^{(l)}$ and $m^{(l)}$ denote the channel number and size (height times width) of hidden features in the lth layer, respectively. For HSI input, each element in Gram matrix G can be expressed as

$$G_{i,j}(H) = \sum_{b=1}^{M} H_{i,b} H_{j,b} \tag{1}$$

where i and j are the channel indices in HSI and $G_{i,j}$ indicates the structure relationship of bands i and j. Specifically, $G_{i,j}$ is calculated by the inner product of the vectorized feature maps of the ith and the jth channels in HSI. Since the Gram matrix reflects information about features being activated together, spectral-structure relations within HSI can be well captured through using $G_{i,j}(H)$ (spectral-structure Gram matrix in Fig. 1). Similarly, $G_{i,j}(f^{(l)}(H, L))$ can be

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obtained to characterize channel structure of the fusion block. On the other side, for texture-structure acquisition of LiDAR data, $\hat{G}_{i,j}(L)$ (texture-structure Gram matrix) can be calculated by the inner product of the ith and the jth vectorized pixels in LiDAR data. Similarly, each element in $\hat{G}_{i,j}$ can be expressed as

$$\hat{G}_{i,j}(L) = \sum_{b=1}^{D} L_{i,b} L_{j,b} \tag{2}$$

which indicates the structure relationship of pixels i and j. Accordingly, $\hat{G}_{i,j}(f^{(l)}(H, L))$ is the texture Gram matrix of the fusion block at the lth layer. Apparently, the structure differences between the fusion block and specific reference source can be measured by the following structural metrics, and the loss terms for these metrics are defined as

$$\begin{cases} L_{\text{Gram}}^{\text{Spec}} = \dfrac{1}{4\hat{d}^{(l)^2} m^{(l)^2}} \displaystyle\sum_{i=1}^{\hat{d}^{(l)}} \sum_{j=1}^{\hat{d}^{(l)}} \left(G_{i,j}(H) - G_{i,j}(f^{(l)}(H, L))\right)^2 \\ L_{\text{Gram}}^{\text{Spat}} = \dfrac{1}{4\hat{d}^{(l)^2} m^{(l)^2}} \displaystyle\sum_{i=1}^{m^{(l)}} \sum_{j=1}^{m^{(l)}} \left(\hat{G}_{i,j}(L) - \hat{G}_{i,j}(f^{(l)}(H, L))\right)^2 \end{cases} \tag{3}$$

where $L_{\text{Gram}}^{\text{Spec}}$ and $L_{\text{Gram}}^{\text{Spat}}$ denote the his- and LiDAR-guided structure losses, respectively. Note that the channel number of the fusion block is the same as that of HSI, and the spatial size of feature maps is the same as that of HSI- and LiDAR-input, i.e., $\hat{d}^{(l)} = D$, $m^{(l)} = M$. Let R^H and R^L be the reconstructed HSI and LiDAR data, respectively. The mean square error (MSE) loss is used for the optimization of multisource data reconstruction task, which is defined as

$$\begin{cases} L_{\text{Recons}}^{H} = \dfrac{1}{M} \displaystyle\sum_{i=1}^{M} \left(H_i - R_i^H\right)^2 \\ L_{\text{Recons}}^{L} = \dfrac{1}{M} \displaystyle\sum_{i=1}^{M} \left(L_i - R_i^L\right)^2. \end{cases} \tag{4}$$

The overall training loss of the proposed IP-CNN is the combination of weighted structure control loss and the MSE loss

$$L = \lambda\left(L_{\text{Gram}}^{\text{Spec}} + L_{\text{Gram}}^{\text{Spat}}\right) + L_{\text{Recons}}^{H} + L_{\text{Recons}}^{L} \tag{5}$$

where λ is a balance parameter. The Gram constrained metrics can be considered as a form of regularization, and the reconstruction setting can help to maintain the fusion information integrity of HSI and LiDAR data. Accordingly, λ has to be adjusted such that the information fusion process is not overregularized.

B. Classification and Training Strategy

As shown in Fig. 1, the fusion block derived from the CSC module is connected to the two-branch CNN [25], and the detailed structure of the two-branch CNN [25] is shown in Fig. 3. In particular, the fusion block is connected with the original HSI-related branches of the two-branch CNN, while HSI data are fed into the LiDAR-related branch. Since the fusion block is acquired by following the guidance of multiple structural constraints, some details of reference data are missing inevitably in the higher layers, so HSI data are reused to further ensure the integrity of information. In particular, the cascade block is retained here to combine diverse features from unequal layers for HSI-feature reuse and propagation. Specifically, the cascade block consists of seven layers of operations, including convolution, batch normalization, and leaky ReLU. In the joint classification step, the data-driven fusion result can be well adjusted by the two-branch CNN, and the task-specific adaption process can achieve satisfying classification performance.

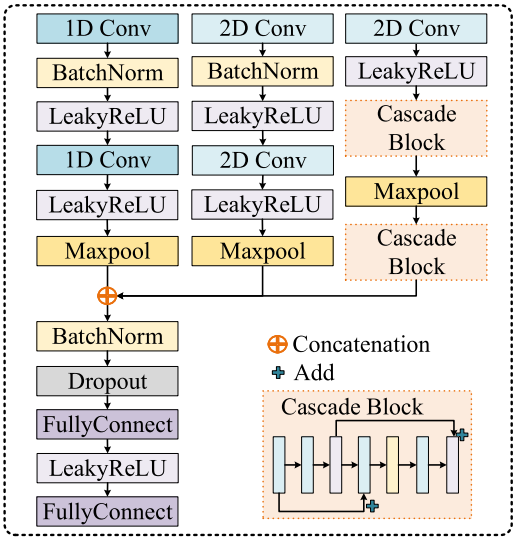


Fig. 3. Architecture of the joint classification model.

All the parameters involved in the proposed framework need to be trained, and the training strategy is listed in Algorithm 1. In the first stage, training patches of HSI and LiDAR data are collected by adopting a sliding window over the same area. Without label restriction, the training patches are sufficient for parameter adjustment in the IP-CNN. In particular, all the convolution operations are executed with zero padding, and the convolution stride is set as 1. The IP-CNN is trained from scratch using the training split with an initial learning rate of 0.001. The learning process is implemented following the policy of Adam [41] for 500 epochs. Moreover, in the second stage, the IP-CNN is adjusted together with the two-branch CNN optimization. Specifically, the pretrained IP-CNN weights are reloaded, and the fusion layer of the IP-CNN is connected to the two-branch CNN. For the two-branch CNN, it still can be seen as training from scratch. Its data setting (three kinds of data augmentation) and core training details are consistent with [25]. In particular, all weights and biases of the proposed framework are initialized with Glorot normalization, and the input data are all scaled to [0, 1] to accelerate the convergence speed. Specifically, the input data X can be normalized through the formula of $[(X - \text{mean}(X))/\text{std}(X)]$, where $\text{mean}(X)$ and $\text{std}(X)$ represent the mean value and standard deviation value of X, respectively.

III. Experiments and Analysis

Three widely used remote sensing datasets are used to evaluate the proposed IP-CNN framework. All the programs

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TABLE I
DETAILED DESCRIPTION OF THREE MULTISENSOR DATASETS

Dataset	Houston[22]		Trento[10]		MUUFL[42]	
Location	Houston, Texas, USA		Trento, Italy		Long Beach, Mississippi, USA	
Sensor Type	HSI	LiDAR-based DSM	HSI	LiDAR-based DSM	HSI	LiDAR-based DSM
Image Size	349×1905	349×1905	600×166	600×166	325×220	325×220
Spatial Resolution	2.5m	2.5m	1m	1m	$0.54 \times 1.0m$	$0.60 \times 0.78m$
Number of Bands	144	1	63	1	64	2
Wavelength Range	$0.38 - 1.05\mu m$	-	$0.42 - 0.99\mu m$	-	$0.375 - 1.05\mu m$	$1.06\mu m$
Sensor Type	CASI-1500	-	AISA Eagle	Optech ALTM 3100EA	CASI-1500	Gemini ALTM LIDAR

Algorithm 1 Training Strategy for Joint Classification of HSI and LiDAR Data

1. Stage 1:
2. Initialize all weights of IP-CNN
3. **While** *epoch* $<$ *epochs* **do**
   - *Train the IP-CNN*
4. **End while**
5. Stage 2:
6. Initialize all weights of two-branch CNN and load weights of IP-CNN
7. **While** *epoch* $<$ *epochs* **do**
   - *Train the two-branch CNN and finetune IP-CNN*
8. **End while**

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE MUUFL DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Trees	150	23246
2	Mostly grass	150	4270
3	Mixed ground surface	150	6882
4	Dirt and sand	150	1826
5	Road	150	6687
6	Water	150	466
7	Building Shadow	150	2233
8	Building	150	6240
9	Sidewalk	150	1385
10	Yellow curb	150	183
11	Cloth panels	150	269
Total		1650	53687

TABLE III
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE TRENTO DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Apple trees	129	4034
2	Buildings	125	2903
3	Ground	105	479
4	Woods	154	9123
5	Vineyard	184	10501
6	Roads	122	3174
Total		819	30214

TABLE IV
NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON DATA

Class		Number of Samples	
No.	Name	Train	Test
1	Health grass	198	1251
2	Stressed grass	190	1254
3	Synthetic grass	192	697
4	Tress	188	1244
5	Soil	186	1242
6	Water	182	325
7	Residential	196	1268
8	Commercial	191	1244
9	Road	193	1252
10	Highway	191	1227
11	Railway	181	1235
12	Parking lot 1	192	1233
13	Parking lot 2	184	469
14	Tennis court	181	428
15	Running track	187	660
Total		2832	15029

are conducted using Keras[1] with Intel Core i7 8700 and with Nvidia 1660 GPUs. Keras is the high-level application programming interface of Tensorflow,[2] and Tensorflow is an open-source software library for numerical computation and machine intelligence using data flow graphs.

A. Experimental Data

The detailed data descriptions are shown in Table I.

1) Muufl Gulfport (MUUFL) Data [42]: This dataset was acquired on November 8, 2010, over the University of Southern Mississippi Gulf Park Campus. The data collection contains coregistered hyperspectral and LiDAR data over the campus. The CASI-1500 HSI scene consists of 64 spectral bands with wavelength ranging from 375 to 1050 nm, including 11 classes. The spatial resolution is 0.54 m across track and 1.0 m along track. Table II lists the number of training and testing samples.

2) Trento Data [10]: The second scene was collected over a rural area in the south of the city of Trento, Italy. The data are composed of 600×166 pixels covering six classes with a spatial resolution of 1 m. The HSI data, which were captured by the AISA Eagle sensor, comprise 63 spectral bands covering the range from 0.42 to 0.99 μm. The LiDAR data were acquired by the Optech ALTM 3100EA sensor. Table III lists the available training and testing samples.

3) Houston Data [22]: The third scene was acquired in 2012 by an airborne sensor over the area of University of Houston campus and neighbor area. The data consist of 349×1905 pixels with a spatial resolution of 2.5 m. The hyperspectral data were acquired on June 23, 2012. The average height of the sensor above ground was 5500 ft. The HSI scene consists of 144 spectral bands with a wavelength

[1]https://github.com/fchollet/keras
[2]http://tensorflow.org/

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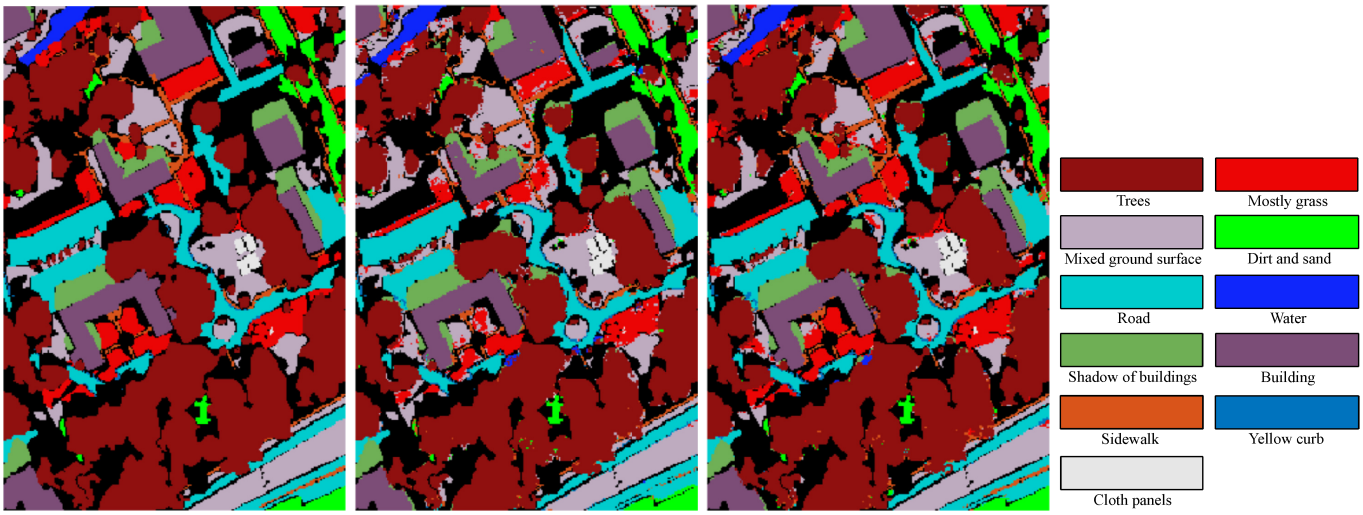


Fig. 4. Dataset visualization and classification maps for the MUUFL data obtained with different methods including: (a) Ground truth map, (b) two-branch CNN (Merge) (91.95%), and (c) IP-CNN (93.83%).

TABLE V
CLASSIFICATION PERFORMANCE OF INFORMATION FUSION WITH DIFFERENT MODULE SETTING ON MUUFL DATA

Different Fusion Setting		Without Gram Matching	Without Reconstruction	IP-CNN
1	Trees	94.30±0.06	93.97±0.25	94.40±0.16
2	Mostly grass	91.28±0.07	91.77±0.18	92.26±0.23
3	Mixed ground surface	88.10±0.48	87.25±0.38	87.96±0.28
4	Dirt and sand	97.27±0.04	97.67±0.21	97.15±0.19
5	Road	93.80±0.24	93.60±0.10	94.38±0.13
6	Water	99.85±0.10	99.75±0.09	99.79±0
7	Building Shadow	96.21±0.48	96.16±0.24	96.30±0.35
8	Building	95.82±0.23	95.84±0.17	96.13±0.09
9	Sidewalk	93.00±0.63	93.18±0.67	94.01±0.29
10	Yellow curb	99.84±0.25	99.45±0	100.00±0
11	Cloth panels	99.63±0	99.67±0.11	99.63±0
OA (%)		93.62±0.19	93.40±0.15	**93.86±0.10**
AA (%)		95.37±0.15	95.30±0.10	**95.64±0.06**
Kappa Coefficient (%)		91.68±0.24	91.40±0.19	**91.99±0.12**

TABLE VI
CLASSIFICATION PERFORMANCE OF INFORMATION FUSION WITH DIFFERENT MODULE SETTINGS ON TRENTO DATA

Different Fusion Setting		Without Gram Matching	Without Reconstruction	IP-CNN
1	Apple trees	99.50±0.19	98.87±0.28	99.00±0.24
2	Buildings	99.65±0.19	99.26±0.16	99.40±0.21
3	Ground	98.60±0.54	97.60±0.44	99.10±0.56
4	Woods	98.91±0.32	99.78±0.30	99.92±0.11
5	Vineyard	99.76±0.06	99.66±0.06	99.66±0.05
6	Roads	78.26±1.52	81.49±1.25	90.21±0.61
OA (%)		97.08±0.38	97.52±0.37	**98.58±0.14**
AA (%)		95.78±0.22	96.11±0.20	**97.88±0.10**
Kappa Coefficient (%)		96.47±0.63	96.81±0.20	**98.17±0.07**

ranging from 0.38 to 1.05 μm, including 15 classes. The LiDAR data were acquired on June 22, 2012. The average height of the sensor above ground was 2000 ft. Table IV lists the available number of training and testing samples.

B. Parameter Tuning and Module Analysis

The classification performance is closely related to the designed architecture of the deep network. In order to validate the proposed method, the classification results using different structure configurations are compared.

1) Impact of Gram Matching and Reconstruction: Joint control of all kinds of constraints can achieve the complementary structure maintenance and fusing information integrity for the HSI and LiDAR source. The performance of "IP-CNN without Gram matching" and "IP-CNN without reconstruction" is first studied. For IP-CNN, the balance parameter is set as $\lambda = 10^{-2}$. As listed in Tables V–VII, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are all studied for performance evaluation. The experiments are implemented ten times, and the related results with average

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TABLE VII
CLASSIFICATION PERFORMANCE OF INFORMATION FUSION WITH DIFFERENT MODULE SETTING ON HOUSTON DATA

Different Fusion Setting		Without Gram Matching	Without Reconstruction	IP-CNN
1	Health grass	85.44±0.15	85.77±0	85.77±0
2	Stressed grass	87.31±0.13	86.58±0.38	87.34±0.17
3	Synthetic grass	100.00±0	100.00±0	100.00±0
4	Tress	94.23±0.16	98.55±1.10	94.26±0.09
5	Soil	98.82±0.60	98.86±0.49	98.42±0.44
6	Water	100.00±0	100.00±0	99.91±0.20
7	Residential	88.66±0.60	90.82±0.77	94.59±1.16
8	Commercial	80.88±1.39	88.23±2.30	91.81±1.90
9	Road	90.62±0.73	88.10±0.66	89.35±0.84
10	Highway	72.30±0.12	68.48±1.02	72.43±0.44
11	Railway	84.70±0.72	92.39±0.66	96.57±1.77
12	Parking lot 1	96.49±0.87	95.54±1.12	95.60±0.81
13	Parking lot 2	93.84±0.77	93.84±0.46	94.37±0.60
14	Tennis court	99.77±0.23	99.95±0.09	99.86±0.15
15	Running track	99.97±0.06	100.00±0	99.99±0.05
OA (%)		89.83±0.13	90.98±0.35	**92.06±0.17**
AA (%)		91.53±0.13	92.47±0.27	**93.35±0.14**
Kappa Coefficient (%)		89.05±0.14	90.27±0.38	**91.42±0.18**

TABLE VIII
CLASSIFICATION PERFORMANCE OF THE PROPOSED METHOD WITH DIFFERENT PATCH SCALES

	Metrics	9×9	11×11	13×13	15×15
MUUFL	OA (%)	93.04	93.45	93.86	93.49
	AA (%)	94.58	95.38	95.64	95.38
	Kappa	90.93	91.46	91.99	91.52
Trento	OA (%)	97.94	98.68	98.58	97.12
	AA (%)	97.31	98.02	97.88	94.75
	Kappa	97.25	98.24	98.17	96.15
Houston	OA (%)	89.63	89.59	92.06	91.19
	AA (%)	91.37	91.25	93.35	92.52
	Kappa	88.85	88.78	91.42	90.50

and standard deviation metrics are reported here. When the information fusion process is implemented without Gram matching, the incomplete information surveillance decreases the ability of feature regulation, leading to the decreased classification performance. Similarly, the information fusion process that removes the reconstruction setting also reduces the ability of fusion information control and results in lower joint classification performance. Based on the comparisons shown in Tables V–VII, we observe that the information fusion guided with both Gram matching and reconstruction setting provides the best overall performance.

2) Multiscale Comparison: The effect of different sizes of input patch is then studied. The performance on different window sizes of 9×9, 11×11, 13×13, and 15×15 is listed in Table VIII. Specifically, the results indicate that the input blocks with different sizes yield different classification performances. Also, the input size of 13×13 offers the best performance for MUUFL data and Houston data, whereas the input sizes of 11×11 and 13×13 achieve comparative classification performance when using Trento data. Accordingly, the input size of 13×13 is appropriate for all datasets.

3) Impact of Information Fusion: In the designed IP-CNN, the data-driven information processing enables seamless multisource information integration, while the classification procedure based on the two-branch CNN can adaptively minimize the gap between fusion information and semantic goal. Experiments are conducted to evaluate the impact of information fusion and semantic adaption separately. For information fusion module evaluation, the experiments are conducted in two respects: 1) IP-CNN derived fusion block is replaced with the original data sources to generate two-branch CNN-based classification results and 2) the performance of the information fusion block derived from the IP-CNN is compared with fusion block obtained by GS-merge [31]. As listed in Table IX, if we remove the IP-CNN-based information fusion and replace it with the original multisource inputs, there would be obviously performance degradation. When the information fusion process is replaced by GS-merge, the classification performance of the proposed method decreases clearly as well. Accordingly, it can be demonstrated that the proposed information fusion process based on the IP-CNN is effective for joint classification.

4) Impact of Semantic Adaption With Different Supplementary Sources: In the classification stage, the channel utilization mechanism in [25] has been adjusted for multisource semantic fusion. Specifically, the fusion blocks acquired from the IP-CNN and the original HSI are jointly used to reduce the impact of detailed degradation in information fusion, and the LiDAR-related branch in the two-branch CNN has been specifically modified to access the original HSI data. Here, the selection of the supplementary source lies in two aspects: 1) HSI is the data source with much larger information capacity than LiDAR source and 2) both spatial and spectral knowledge are components of HSI data, and thus, HSI data bear a stronger resemblance with the fusion block than LiDAR data. Therefore, HSI data are used as the supplementary source, and the data volume difference and the distribution discrepancy difference are smaller, further improving the classification performance. As listed in Table X, where "Fusion-H" and "Fusion-L" mean the classification procedure using the fusion block coupled with HSI data and LiDAR data, respectively. Besides, let "Fusion" denote the classification procedure based on the fusion block with HSI-channel classifier. Compared with Table IX, even without supplementary design, the performance of "Fusion" is already better than that of the existing two-branch CNN. For MUUFL data and Houston data, when the supplementary source is LiDAR data, the

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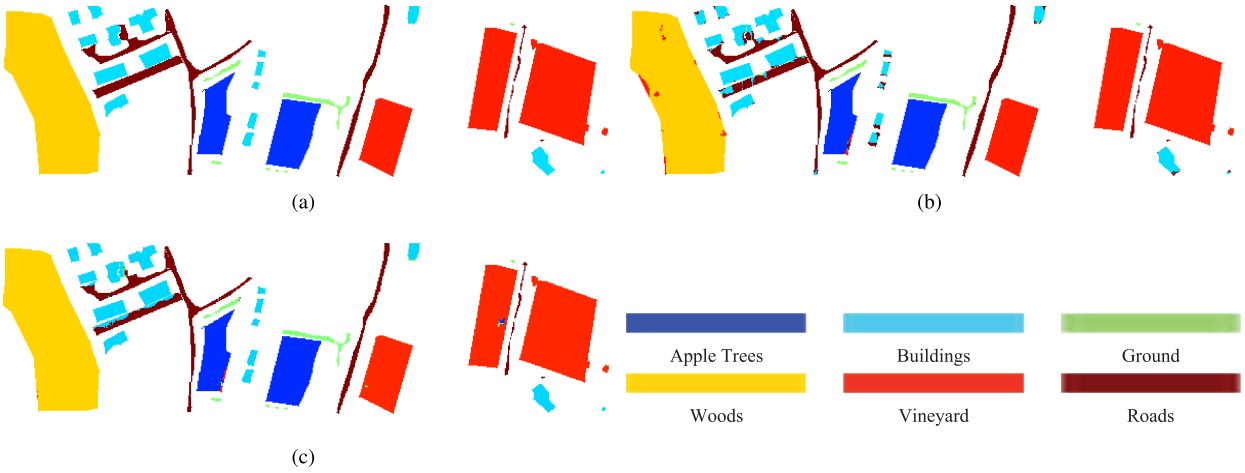


Fig. 5. Dataset visualization and classification maps for the Trento data obtained with different methods including: (a) Ground truth map, (b) CNN-MRF (98.40%), and (c) IP-CNN (98.65%).

TABLE IX
COMPARISON OF THE PROPOSED CLASSIFICATION FRAMEWORK WITH DIFFERENT MODULE CONFIGURATIONS

Methods			Metrices	Datasets		
GS-Merge	IP-CNN	two-branch CNN		MUUFL	Trento	Houston
×	×	✓	OA(%)	90.35	97.92	87.98
			AA(%)	91.27	96.19	90.11
			Kappa	87.27	96.81	86.98
✓	×	✓	OA(%)	91.95	97.46	88.91
			AA(%)	92.58	96.80	90.42
			Kappa	89.35	96.61	87.96
×	✓	✓	OA(%)	93.86	98.58	92.06
			AA(%)	95.64	97.88	93.35
			Kappa	91.99	98.17	91.42

TABLE X
PERFORMANCE OF SEMANTIC ADAPTION WITH DIFFERENT SUPPLEMENTARY SOURCES

	Metrics	Fusion	Fusion-L	Fusion-H
MUUFL	OA (%)	92.44	93.43	93.86
	AA (%)	94.47	95.07	95.64
	Kappa	90.16	91.42	91.99
Trento	OA (%)	98.41	97.47	98.58
	AA (%)	97.39	95.93	97.88
	Kappa	97.88	96.63	98.17
Houston	OA (%)	91.33	91.98	92.06
	AA (%)	92.78	93.28	93.35
	Kappa	90.63	91.35	91.42

classification performance of "Fusion-L" is better than that of "Fusion," while on Trento data, the classification performance using LiDAR supplementary obviously decreased. However, the proposed semantic adaption process, "Fusion-H," improves the performance by a large margin on the three datasets.

C. Classification Performance and Analysis

To validate the effectiveness, the proposed method is compared with several competitive classifiers, such as recently developed CNN-PPF [14], two-branch CNN [25], FDSSCN [17], context CNN [15], CRNN [16], CNN-MRF [43], IAP [44], Coupled CNN [32], and Semi-GAN. Note that all the comparative methods are implemented with optimal parameters. For the two-branch CNN (Merge), the HSI-channel input is replaced with data fused from the GS-merge strategy. For the Semi-GAN, the interleaving perception network is integrated with the generative adversarial protocol. Specifically, in Semi-GAN, the generative part is the same as the original IP-CNN, and the discriminator is added to construct the adversarial process. The network configuration of the discriminator in Semi-GAN is the same as the one in [45], and it is utilized to determine whether the data reconstructed from the generator are true or false.

Tables XI–XIII list the OA, AA, and Kappa Coefficient (Kappa) for three experimental data. The proposed method is obviously superior to other methods. Specifically, using the MUUFL dataset for evaluation, it shows approximately 2.89%, 3.51%, 9.03%, 1.91%, 7.79%, 4.92%, 2.48%, 7.81%, and 2.93%, 2% improvements over CNN-PPF, two-branch CNN, FDSSCN, two-branch CNN (Merge), context CNN, CNN-MRF, CRNN, IAP, Coupled CNN, and Semi-GAN, respectively. For the Trento dataset, it yields approximately 3.82%, 0.66%, 10.38%, 1.12%, 2.47%, 0.18%, 1.36%, 2.28%, 0.89%, and 1.38% improvements over CNN-PPF, two-branch CNN, FDSSCN, two-branch CNN (Merge), context CNN, CNN-MRF, CRNN, IAP, Coupled CNN, and Semi-GAN, respectively. Furthermore, for the Houston dataset, its improvement is about 8.73%,

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TABLE XI

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE MUUFL DATA

No.	Class	Performance										
		CNN-PPF	two-branch CNN	FDSSCN	two-branch CNN (Merge)	context CNN	CNN-MRF	CRNN	IAP	Coupled CNN	Semi-GAN	IP-CNN
1	Trees	89.07	92.35	87.37	94.28	91.29	93.04	91.43	85.32	98.90	92.85	94.40
2	Mostly grass	85.71	59.30	32.37	82.84	63.09	60.17	63.16	81.99	78.60	89.16	92.26
3	Mixed ground surface	80.15	94.47	88.12	84.58	81.84	90.60	90.20	78.51	90.66	82.26	87.96
4	Dirt and sand	93.10	93.74	94.51	97.32	93.92	97.20	93.44	94.63	90.60	96.60	97.15
5	Road	88.98	92.76	97.84	92.96	89.44	92.00	87.62	86.81	96.90	91.89	94.38
6	Water	98.93	98.42	96.20	99.05	92.92	99.68	95.89	99.79	75.98	100.00	99.79
7	Building shadow	89.07	95.68	89.92	93.28	84.73	95.39	90.16	90.91	73.54	96.78	96.30
8	Building	92.15	94.01	87.44	95.42	81.22	94.71	89.29	95.46	96.66	95.87	96.13
9	Sidewalk	75.45	86.64	85.75	84.13	81.30	30.53	82.91	73.94	64.93	93.65	94.01
10	Yellow curb	100.00	100.00	72.73	100.00	98.91	36.36	96.97	98.91	19.47	99.45	100.00
11	Cloth panels	100.00	96.64	99.16	97.48	99.63	95.80	96.64	99.63	62.76	99.63	99.63
	OA	90.97	90.35	84.83	91.95	86.07	88.94	91.38	86.05	90.93	91.86	**93.86**
	AA	90.24	91.27	84.70	92.58	87.12	85.02	88.88	89.63	77.18	94.38	**95.64**
	Kappa	84.46	87.27	80.24	89.35	81.89	85.55	84.41	82.12	88.22	89.43	**91.99**

TABLE XII

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE TRENTO DATA

No.	Class	Performance										
		CNN-PPF	two-branch CNN	FDSSCN	two-branch CNN(Merge)	context CNN	CNN-MRF	CRNN	IAP	Coupled CNN	Semi-GAN	IP-CNN
1	Apple trees	90.11	98.07	67.97	98.51	99.26	99.95	97.72	96.26	99.87	98.07	99.00
2	Buildings	83.34	95.21	98.31	92.49	86.81	89.97	95.69	97.80	83.84	97.28	99.40
3	Ground	71.13	93.32	66.39	100.00	97.91	98.33	100.00	99.79	87.09	100.00	99.10
4	Woods	99.04	99.93	89.60	97.32	97.31	100.00	96.85	95.67	99.98	99.64	99.92
5	Vineyard	99.37	98.78	100.00	100.00	99.82	100.00	100.00	98.19	99.61	99.73	99.66
6	Roads	89.73	89.98	64.93	92.56	84.63	93.98	77.76	90.01	98.75	80.25	90.21
	OA	94.76	97.92	88.20	97.46	96.11	98.40	97.22	96.30	97.69	97.20	**98.58**
	AA	88.97	96.19	81.20	96.80	94.29	97.04	94.67	96.29	94.86	95.83	**97.88**
	Kappa	93.04	96.81	83.94	96.61	94.81	97.86	96.29	95.07	96.91	96.27	**98.17**

TABLE XIII

COMPARISON OF THE CLASSIFICATION ACCURACY (%) USING THE HOUSTON DATA

No.	Class	Performance										
		CNN-PPF	two-branch CNN	FDSSCN	two-branch CNN(Merge)	context CNN	CNN-MRF	CRNN	IAP	Coupled CNN	Semi-GAN	IP-CNN
1	Health grass	83.57	83.10	85.53	83.10	84.89	85.77	83.00	84.57	98.51	85.77	85.77
2	Stressed grass	98.21	84.10	84.61	81.20	87.40	86.28	79.41	87.40	97.83	87.40	87.34
3	Synthetic grass	98.42	100.00	99.43	100.00	99.86	99.00	99.80	99.57	70.60	100.00	100.00
4	Tress	97.73	93.09	94.13	92.90	93.49	92.85	90.15	92.68	99.06	99.28	94.26
5	Soil	96.50	100.00	100.00	99.81	100.00	100.00	99.71	99.92	100.00	99.19	98.42
6	Water	97.20	99.30	98.15	100.00	98.77	98.15	83.21	98.15	41.11	100.00	99.91
7	Residential	85.82	92.82	92.11	92.54	82.81	91.64	88.06	87.54	83.14	89.27	94.59
8	Commercial	56.51	82.34	65.03	94.87	78.78	80.79	88.61	80.63	98.39	84.08	91.81
9	Road	71.20	84.70	66.21	83.85	82.51	91.37	66.01	86.50	94.81	90.42	89.35
10	Highway	57.12	65.44	71.88	69.89	59.41	73.35	52.22	99.10	92.98	68.05	72.43
11	Railway	80.55	88.24	84.53	86.15	83.24	98.87	81.97	82.67	90.88	97.89	96.57
12	Parking lot 1	62.82	89.53	98.54	92.60	92.13	89.38	69.83	96.92	91.02	91.00	95.60
13	Parking lot 2	63.86	92.28	90.41	79.30	94.88	92.75	79.64	85.71	97.09	94.88	94.37
14	Tennis court	100.00	96.76	100.00	100.00	99.77	100.00	100.00	99.77	100.00	100.00	99.86
15	Running track	98.10	99.79	100.00	100.00	98.79	100.00	100.00	99.39	97.85	98.64	99.99
	OA	83.33	87.98	86.60	88.91	86.90	90.61	88.55	90.98	90.43	90.88	**92.06**
	AA	83.21	90.11	88.20	90.42	89.11	92.01	90.30	92.04	90.22	92.39	**93.35**
	Kappa	81.88	86.98	85.59	87.96	85.89	89.87	87.56	90.25	89.68	90.16	**91.42**

4.08%, 5.46%, 3.15%, 5.16%, 1.45%, 3.51%, 1.08%, 1.63%, and 1.18% over CNN-PPF, two-branch CNN, FDSSCN, two-branch CNN (Merge), context CNN, CNN-MRF, CRNN, IAP, Coupled CNN, and Semi-GAN, respectively. Thus, it can be concluded that the proposed information fusion and joint classification obtains better classification accuracy than other comparison baselines due to more effective information utilization. In particular, Semi-GAN tends to pay more attention on multisource data reconstruction rather than the complementary information fusion, which makes final classification less effective than IP-CNN. Also, it is noticeable that all baselines are implemented with the most appropriate input-patch size. The classification performance of different methods with small input patches is further analyzed based on the Houston data, and the related results are shown in Table XIV. When using small patches, the classification performance of different methods decreases obviously compared with the optimal ones reported in Table XIII. Taking CNN-MRF for example,

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TABLE XIV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS WITH SMALL PATCH SCALES ON HOUSTON DATA

Metrics		two-branch CNN	context CNN	CNN-MRF	CRNN	IP-CNN
3×3	OA (%)	83.86	81.48	85.89	87.66	87.51
	AA (%)	86.11	84.49	87.89	88.14	89.55
	Kappa	82.62	80.07	84.76	86.68	86.54
5×5	OA (%)	86.67	86.19	86.04	86.18	88.62
	AA (%)	88.21	88.44	87.26	86.83	90.57
	Kappa	85.63	85.09	84.91	85.10	87.76
7×7	OA (%)	87.47	86.77	90.18	86.57	88.56
	AA (%)	88.10	88.80	91.49	86.71	90.56
	Kappa	86.49	85.72	89.38	85.45	87.68

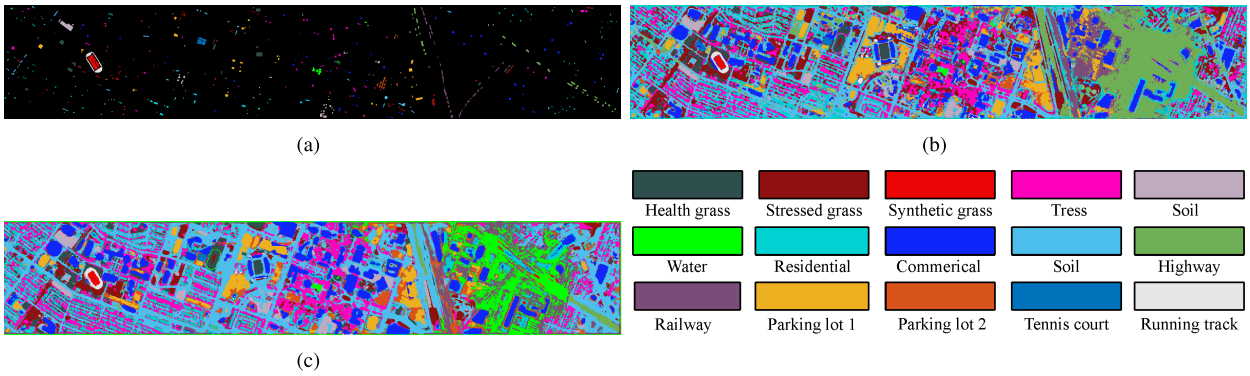


Fig. 6. Dataset visualization and classification maps for the Houston data obtained with different methods including: (a) Ground truth map, (b) IAP (90.98%), and (c) IP-CNN (92.37%).

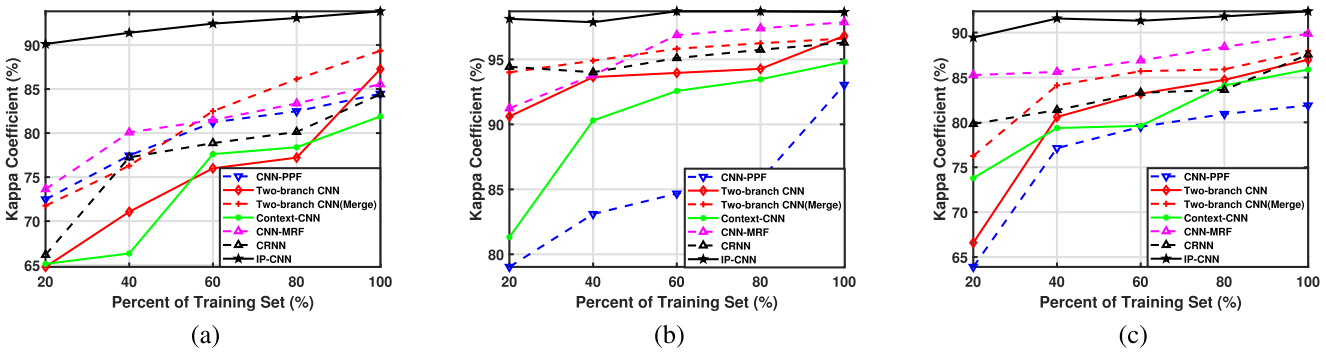


Fig. 7. Classification performance with different sizes of training samples using: (a) MUUFL data, (b) Trento data, and (c) Houston data.

the best OA of CNN-MRF is 90.18% when the patch size is set within the range of 3–7, while in Table XIII, the classification accuracy is 90.61%. Besides, the best performance of the proposed one is also obtained using the larger patch size, as reported in Table XIII. This result also reveals that the reasonable utilization of spatial structure information is of great importance for HSI and LiDAR data joint classification.

Figs. 4–6 show the qualitative results obtained by different classifiers. To facilitate comparison, the ground truth map is also provided. In comparison to the classification results from the most comparative baseline, the proposed IP-CNN obtains visually better results. Taking the MUUFL data for example, the proposed method yields obviously less noisy results, e.g., the mostly grass classes in Fig. 4. In particular, on Houston data, it is obvious that the proposed method offers better classification performance for the Railway and Commercial class in the obscured area.

Fig. 7 shows the performance sensitivity to the training-samples size. The percentage of training samples is changed from 20% to 100%. Here, 100% represents using all training samples, as listed in Tables II–IV. Note that all training samples are selected randomly, and averaged results are reported. Obviously, when compared with other baselines, the IP-CNN obtains better results. When the training set size is small, the proposed information fusion strategy strengthens the ability of joint utilization, providing better classification performance than other baselines. For example, in Fig. 7(a), with 20% training samples, the Kappa coefficient of the IP-CNN is approximately 0.87, whereas the Kappa coefficient of other methods is all below 0.8. The improvement gap is

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obvious, which verifies the robustness of the IP-CNN in small-sample-size situations.

IV. Conclusion

In this article, an information fusion network, called IP-CNN, was proposed to integrate multisource heterogeneous information in an interleaving manner for more accurate joint classification of the HSI and LiDAR data. It integrated data-driven analysis and task-specific adaption into a unified framework, which effectively formed a solution to unsupervised complementary information fusion, reducing the analysis burden of feature adjustment in the classification process. Extensive experiments on three benchmark remote sensing datasets demonstrated that the proposed framework can significantly outperform state-of-the-art methods even with small-size training samples.

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